The Spectral Theory of Metrics, Embeddability and Unstable Sanctions Busters

Based on "The Geometry of Inconvenience and Perverse Equilibria in Trade Networks" (arXiv:2504.07700)

James Eldred Pascoe (Drexel University)

Joint work with **Michael Coopman** (Florida International University),
Austin Jacobs (Santa Fe College), and **Henry Baker Pascoe** (YouTube)

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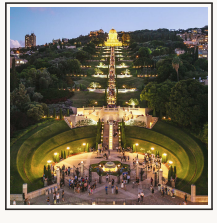
Be'er Sheva / BGU (Dec 25, 2025)



Jerusalem (Dec 28, 2025)



Weizmann Inst. (Dec 22, 2025)



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Visit India! Scenes from the 2025 Two-Month Visit (Bengaluru, Agra & Mumbai)



2025 Two-Month Visit to India: Buland Darwaza & Fatehpur Sikri • Swami Vivekananda at Ramakrishna Math (Bangalore) • Lalbagh Gardens & Flower Show • 79th Independence Day Rangoli • Marine Drive (Mumbai)

1. Why Do Countries Trade?

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2. Geography, Tariffs, and Coercive Economic Sanctions

- **Natural & Tariff Frictions:** Distance, landlocked borders, and tariffs dampen bilateral trade continuously.
- **Sanctions:** A sender coalition (e.g., US/EU) attempts to sever direct trade with targeted states (e.g., Russia, Iran).

Motivation: Trade Frictions, Sanctions, and Sanctions Busters

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3. Sanctions Busters & Bipartite "Anti-Bloc" Networks

- **Crude Oil & Refining Arbitrage:** India and China purchase sanctioned Russian and Iranian crude oil at a discount, refine it domestically, and resell refined petroleum products globally.
- **Transit Hubs:** Turkey, the UAE, and Kazakhstan re-export dual-use goods (paralleling the UAE during Iran sanctions and UK firms during South Africa sanctions), forming **bipartite anti-blocs** that extract intermediary rents.

Visit India! Mathematics, Lectures & Scenes from Bangalore & Kerala (2025 Visit)



2025 Two-Month Visit: ISI Bangalore Guest House • Harmonic Analysis & Representation Theory Lectures • Paradesi Synagogue & Mural Art (Kochi) • Arabian Sea Coast & Kerala Nightlife

Definition & Axioms of Krugman Iceberg Trading Costs

In Krugman's (1980) **iceberg transport model**, if 1 unit of a good is shipped from country i to country j, only a fraction $\phi_{ij} \in (0, 1]$ arrives intact (the remainder $1 - \phi_{ij}$ "melts" in transit).

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Definition (Freeness-of-Trade Matrix Φ)

Let $N \geq 2$ be the number of countries. An $N \times N$ real matrix $\Phi = (\phi_{ij})_{i,j=1}^{N}$ is a **Freeness-of-Trade (FoT) Matrix** if it satisfies the following three axioms for all $1 \leq i, j, k \leq N$:

$$\textbf{(Axiom 1: Normalization \& Positivity)} \quad 0 < \phi_{ij} \leq 1 \quad \text{and} \quad \phi_{ii} = 1,$$

$$\textbf{(Axiom 2: No Triangular Arbitrage)} \quad \phi_{ij}\phi_{jk} \leq \phi_{ik},$$

$$\textbf{(Axiom 3: Symmetry)} \quad \phi_{ij} = \phi_{ji}.$$

Economic meaning of Axiom 2: Shipping directly from i to k (ϕ_{ik}) is at least as free as shipping indirectly through any intermediate country j ($\phi_{ij}\phi_{jk}$); otherwise, traders would strictly prefer transshipment via j.

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Remark on the Symmetry Axiom ($\phi_{ij} = \phi_{ji}$)

Real-world tariffs and sanctions may be asymmetric ($\phi_{ij} \neq \phi_{ji}$). However, because we construct a **counterexample** to equilibrium uniqueness *within* the symmetric class, uniqueness **cannot hold in any more general model** either.

Following **Mossay & Tabuchi (2015, *Economic Journal*)**, consider N countries with immobile labor endowments $L_i > 0$ ($\sum_{i=1}^{N} L_i = 1$) and constant elasticity of substitution $\sigma > 1$.

Model Primitives: CES Utility, Mill Pricing, and Price Index

- **Dixit–Stiglitz Utility:** Consumers in country i maximize over varieties $v \in V_j$ ($j = 1, \ldots, N$):

$$U_i = \left(\sum_{j=1}^{N} \int_{V_j} q_{ji}(v)^{\frac{\sigma-1}{\sigma}} \, dv \right)^{\frac{\sigma}{\sigma-1}}.$$

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- **Prices & Firm Mass:** With wage w_i, marginal labor requirement c, fixed cost F, and iceberg factor $r_{ij} \geq 1$ related to freeness of trade by $\phi_{ij} := r_{ij}^{1-\sigma} \in (0, 1]$, monopolistic competition yields

$$p_{ij} = \frac{\sigma c}{\sigma - 1} r_{ij} w_i, \qquad n_i = \frac{L_i}{\sigma F}, \qquad P_i = \left(\sum_{k=1}^{N} n_k p_{ki}^{1-\sigma} \right)^{\frac{1}{1-\sigma}}, \qquad U_i^* = \frac{w_i}{P_i}.$$

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Proposition (Master Equilibrium System)

Setting $v_i := w_i^{\sigma} > 0$ and $\varepsilon := \frac{\sigma}{\sigma-1} > 1$, market clearing ($w_i L_i = \sum_{j=1}^{N} n_i p_{ij} q_{ij}$) reduces to solving $F(v) = 0$ on $\mathbb{R}^N_{>0}$:

$$F_i(v) := v_i - \sum_{j=1}^{N} L_j \phi_{ij} v_j^{-\varepsilon} = 0 \qquad (i = 1, \ldots, N).$$

Why does the 3-country model always admit a unique equilibrium?

Lemma (Mossay & Tabuchi, 2015, Lemma 2: Positive Definiteness for $N = 3$)

Let $\Phi = (\phi_{ij})_{i,j=1}^{3}$ be any non-degenerate 3×3 Freeness-of-Trade matrix ($\phi_{ij} < 1$ for $i \neq j$ and $\phi_{ij}\phi_{jk} < \phi_{ik}$). Then all principal minors of Φ are strictly positive: $1 - \phi_{ij}^2 > 0$, and

$$\det(\Phi) = 1 - \phi_{12}^2 - \phi_{13}^2 - \phi_{23}^2 + 2\phi_{12}\phi_{13}\phi_{23} = \sum_{\text{cyc}} (\phi_{ij} - \phi_{ik}\phi_{jk})(1 - \phi_{ij}) + \prod_{i<j} (1 - \phi_{ij}) > 0.$$

Hence every non-degenerate 3×3 Freeness-of-Trade matrix Φ is **strictly positive definite** ($\Phi \succ 0$).

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Theorem (Mossay & Tabuchi, 2015, Propositions 1–4)

For $N = 3$ countries, for any labor vector $L \in \mathbb{R}^3_{>0}$ and any $\sigma > 1$:

1. **(Existence & Uniqueness)** The equilibrium loci $e_1(w) = 0$ and $e_2(w) = 0$ are strictly monotone inside the admissible wage triangle $\Delta \subset \mathbb{R}^2_{>0}$ with positive Jacobian determinant $\det(E) > 0$, yielding a **unique equilibrium** (w_1^*, w_2^*, w_3^*).

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2. **(Preferential Trade Agreements Harm Third Countries)** A bilateral PTA increasing ϕ_{23} strictly increases relative wages and welfare in partner countries 2 and 3, while strictly reducing welfare $U_1^* = w_1/P_1$ in the excluded country 1.

Problem (Mossay's Open Question)

Must every non-degenerate $N \times N$ Freeness-of-Trade matrix Φ be positive definite? Does the Mossay–Tabuchi model admit a unique equilibrium for arbitrary N?

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Let Φ be an $N \times N$ Freeness-of-Trade matrix. If Φ is positive semi-definite ($\Phi \succeq 0$), then the equilibrium system $F_i(v) = v_i - \sum_{j=1}^{N} L_j \phi_{ij} v_j^{-\varepsilon} = 0$ has a **unique solution** $v^* \in \mathbb{R}^N_{>0}$ for all $L \in \mathbb{R}^N_{>0}$ and all $\varepsilon > 0$.

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Proof (Convex Potential Formulation)

Introduce the change of variables $u_i := L_i v_i^{-\varepsilon} > 0$, so that $v_i = L_i^{1/\varepsilon} u_i^{-1/\varepsilon}$. Then $F_i(v) = 0$ becomes

$$(\Phi u)_i - L_i^{1/\varepsilon} u_i^{-1/\varepsilon} = 0 \qquad (i = 1, \ldots, N).$$

This system is the critical point equation $\nabla V(u) = 0$ on the open positive orthant $\mathbb{R}^N_{>0}$ for the scalar potential

$$V(u) := \frac{1}{2}\langle \Phi u, u \rangle - \frac{\varepsilon}{\varepsilon - 1} \sum_{i=1}^{N} L_i^{1/\varepsilon} u_i^{\frac{\varepsilon-1}{\varepsilon}}, \qquad \nabla^2 V(u) = \Phi + \frac{1}{\varepsilon} \operatorname{diag}\left(L_i^{1/\varepsilon} u_i^{-\frac{1}{\varepsilon}-1} \right).$$

Since the diagonal term is strictly positive definite for every $u \in \mathbb{R}^N_{>0}$, $\Phi \succeq 0 \implies \nabla^2 V(u) \succ 0$ everywhere on the convex domain $\mathbb{R}^N_{>0}$. Thus $V(u)$ is strictly convex on $\mathbb{R}^N_{>0}$ and admits at most one critical point. ■

Definition (Friction-of-Trade Metric Matrix M_Φ)

Given an $N \times N$ Freeness-of-Trade matrix $\Phi = (\phi_{ij})_{i,j=1}^N$, define the **Friction-of-Trade Matrix** $M_\Phi = (m_{ij})_{i,j=1}^N$ entrywise by

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Lemma (Equivalence of FoT Axioms and Metric Space Axioms)

An $N \times N$ matrix $\Phi = (\phi_{ij})$ satisfies the Freeness-of-Trade Axioms (1)–(3) if and only if $M_\Phi = (m_{ij})$ is a **pseudo-metric** on $\{1, \ldots, N\}$:

$$(1')\ \ m_{ij} \geq 0,\ \ m_{ii} = 0; \qquad (2')\ \ m_{ik} \leq m_{ij} + m_{jk}\ \ (\text{since } \phi_{ij}\phi_{jk} \leq \phi_{ik}); \qquad (3')\ \ m_{ij} = m_{ji}.$$

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Definition (Rescaled Family Φ_t and Mossay–Tabuchi Stability)

For $t \in (0, 1)$, rescaling the trade friction metric M_Φ by $s := -\log t > 0$ produces the one-parameter family of Freeness-of-Trade matrices

$$\Phi_t := \left(\phi_{ij}^{-\log t}\right)_{i,j=1}^N = \left(t^{m_{ij}}\right)_{i,j=1}^N = \left(e^{-s m_{ij}}\right)_{i,j=1}^N.$$

We say that Φ (or M_Φ) is **Mossay–Tabuchi (MT) Stable** if $\Phi_t \succeq 0$ for **all** $t \in (0, 1)$.

Definition (Conditionally Negative Semi-Definite Matrix)

An $N \times N$ real symmetric matrix $M = (m_{ij})_{i,j=1}^{N}$ is **Conditionally Negative Semi-Definite (CNSD)** if for every vector $c = (c_1, \ldots, c_N)^T \in \mathbb{R}^N$ satisfying $\sum_{i=1}^{N} c_i = 0$ (i.e., $c \in \mathbf{1}^{\perp}$),

$$\langle Mc, c \rangle = \sum_{i=1}^{N} \sum_{j=1}^{N} m_{ij} c_i c_j \leq 0.$$

Equivalently, if $P = I_N - \frac{1}{N}\mathbf{1}\mathbf{1}^T$ is orthogonal projection onto $\mathbf{1}^{\perp}$, then M is CNSD if and only if $-PMP \succeq 0$.

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Theorem 4.1 (Schoenberg, 1938; Characterization of MT Stability)

Let Φ be an $N \times N$ Freeness-of-Trade matrix with friction metric $M_\Phi = (m_{ij})$. The following are equivalent:

1. Φ is **Mossay–Tabuchi stable**, i.e., $\Phi_t = (e^{-s m_{ij}})_{i,j=1}^N \succeq 0$ for all $s = -\log t > 0$.
2. The friction metric M_Φ is **Conditionally Negative Semi-Definite (CNSD)**.
3. **(Negative-Type Euclidean Embedding)** There exist points $x_1, \ldots, x_N \in \mathbb{R}^N$ such that

$$m_{ij} = \|x_i - x_j\|_2^2 \qquad \text{for all } 1 \leq i, j \leq N.$$

Proof of (1) $\implies$ (2)*:* Fix $c \in \mathbf{1}^\perp$. Then $f(s) := \langle \Phi_{e^{-s}} c, c \rangle = \sum_{i,j} e^{-s m_{ij}} c_i c_j \geq 0$ for all $s > 0$, with $f(0) = (\sum_i c_i)^2 = 0$. Differentiating at $s = 0^+$ yields $0 \leq f'(0) = -\sum_{i,j} m_{ij} c_i c_j$, so $\sum_{i,j} m_{ij} c_i c_j \leq 0$. ■

Why does Schoenberg's theorem equate Mossay–Tabuchi stability with **squared** Euclidean distance $m_{ij} = \|x_i - x_j\|_2^2$ rather than ordinary Euclidean distance $\|x_i - x_j\|_2$?

The "One-Day Shipping" Model of Fragile Goods (e.g., Ceramic Plates)

Suppose countries sit at points $x_1, \ldots, x_N$ in an ℓ^2 space and must ship fragile goods (such as plates) from country i to country j within a **fixed delivery window** $T = 1$ ("one-day shipping"):

1. **Required Transit Speed:** To cover the Euclidean distance $d_{ij} = \|x_i - x_j\|_2$ in fixed time $T = 1$, the shipment must travel at velocity

$$v_{ij} = \frac{\|x_i - x_j\|_2}{T} = \|x_i - x_j\|_2.$$

Physical Interpretation of Schoenberg's Theorem: "One-Day Shipping Metrics"

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$$v_{ij} = \frac{\|x_i - x_j\|_2}{T} = \|x_i - x_j\|_2.$$

2. **Breakage Proportional to Speed Along the Route:** If fragile plates break at a rate per unit distance proportional to the speed v_{ij} at which they are rushed (or equivalently, the kinetic/vibrational energy dissipated over the trip is $\mathcal{E}_{ij} = \frac{1}{2}\kappa v_{ij}^2 \propto \|x_i - x_j\|_2^2$), then along the path $x \in [0, d_{ij}]$ the surviving fraction $q(x)$ satisfies

$$\frac{dq}{dx} = -s v_{ij}\, q(x) \quad \Longrightarrow \quad \phi_{ij} = \frac{q(d_{ij})}{q(0)} = \exp(-s v_{ij} d_{ij}) = \exp\!\left(-s\|x_i - x_j\|_2^2\right) = e^{-s m_{ij}}.$$

Physical Interpretation of Schoenberg's Theorem: "One-Day Shipping Metrics"

Why does Schoenberg's theorem equate Mossay–Tabuchi stability with **squared** Euclidean distance $m_{ij} = \|x_i - x_j\|_2^2$ rather than ordinary Euclidean distance $\|x_i - x_j\|_2$?

The "One-Day Shipping" Model of Fragile Goods (e.g., Ceramic Plates)

Suppose countries sit at points $x_1, \ldots, x_N$ in an ℓ^2 space and must ship fragile goods (such as plates) from country i to country j within a **fixed delivery window** $T = 1$ ("one-day shipping"):

1. **Required Transit Speed:** To cover the Euclidean distance $d_{ij} = \|x_i - x_j\|_2$ in fixed time $T = 1$, the shipment must travel at velocity

$$v_{ij} = \frac{\|x_i - x_j\|_2}{T} = \|x_i - x_j\|_2.$$

2. **Breakage Proportional to Speed Along the Route:** If fragile plates break at a rate per unit distance proportional to the speed v_{ij} at which they are rushed (or equivalently, the kinetic/vibrational energy dissipated over the trip is $\mathcal{E}_{ij} = \frac{1}{2}\kappa v_{ij}^2 \propto \|x_i - x_j\|_2^2$), then along the path $x \in [0, d_{ij}]$ the surviving fraction $q(x)$ satisfies

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Economic Meaning of CNSD Metrics via Schoenberg's Theorem

A Freeness-of-Trade matrix Φ is **Mossay–Tabuchi stable** ($\Leftrightarrow$ M_Φ is CNSD) if and only if there exists an ℓ^2 space in which the friction metric $m_{ij} = -\log \phi_{ij} = \|x_i - x_j\|_2^2$ is the **cost to get from point A (x_i) to point B (x_j) in a fixed amount of time**.

Proposition (CNSD Metrics Form a Closed Convex Cone)

If M_1, M_2 are CNSD metrics on N points and $\alpha, \beta \geq 0$, then $\alpha M_1 + \beta M_2$ is a CNSD metric.

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Definition & Lemma (Every Cut Metric Is CNSD)

For any non-empty proper subset $S \subset \{1, \ldots, N\}$, the **cut metric** δ_S is defined by $\delta_S(i,j) := |1_S(i) - 1_S(j)| = |1_S(i) - 1_S(j)|^2 \in \{0, 1\}$. For every vector $c \in \mathbb{R}^N$ with $\sum_{i=1}^N c_i = 0$,

$$\sum_{i,j=1}^{N} \delta_S(i,j) c_i c_j = \sum_{i,j=1}^{N} \big(1_S(i) + 1_S(j) - 2 \cdot 1_S(i) 1_S(j)\big) c_i c_j = -2 \left(\sum_{i \in S} c_i \right)^2 \leq 0.$$

Hence **every cut metric is CNSD** (and therefore Mossay–Tabuchi stable).

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Hence **every cut metric is CNSD** (and therefore Mossay–Tabuchi stable).

Corollary (Every ℓ^1-Embeddable Metric Is CNSD and MT-Stable)

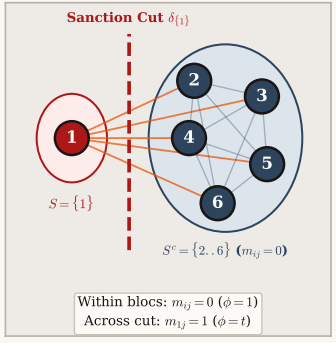
A finite metric space (X, M) embeds isometrically into ℓ^1 if and only if M lies in the conical hull of cut metrics, $\mathrm{CUT}_N := \mathrm{cone}\{\delta_S : S \subset \{1, \ldots, N\}\}$. Since CNSD_N is a convex cone containing every cut metric,

$$\mathrm{CUT}_N(\ell^1) \subset \mathrm{CNSD}_N(\ell_2^2) \subset \mathrm{MET}_N,$$

so **every ℓ^1-embeddable trade metric is CNSD, MT-stable, and admits a unique equilibrium**.

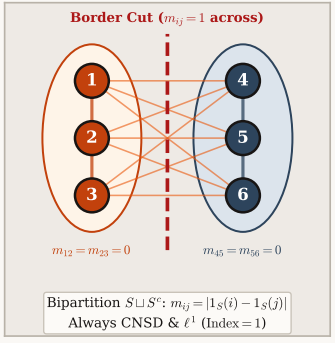
Visualizing Cut Metrics ($C_{k,N-k}$), Tree Metrics, and ℓ^1 Geometry

1. Singleton Cut $C_{1,5} = \delta_{\{1\}}$



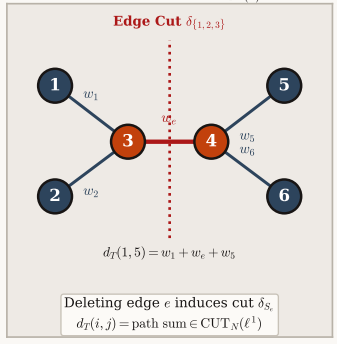
Within blocs: $m_{ij} = 0$ ($\phi = 1$)
Across cut: $m_{1j} = 1$ ($\phi = t$)

2. Split Cut $C_{3,3} = \delta_{\{1,2,3\}}$



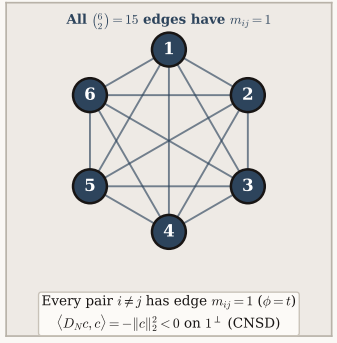
Bipartition $S \sqcup S^c$: $m_{ij} = |1_S(i) - 1_S(j)|$
Always CNSD & ℓ^1 (Index $= 1$)

3. Tree Metric $d_T = \sum_{e \in E(T)} w_e \delta_{S_e}$



Deleting edge e induces cut δ_{S_e}
$d_T(i,j) =$ path sum $\in \mathrm{CUT}_N(\ell^1)$

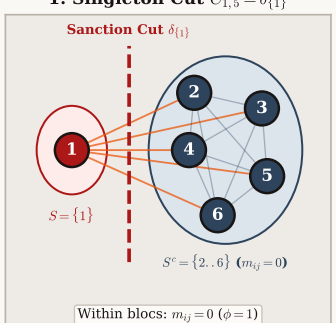
4. Complete Graph K_6 ($D_6 = J_6 - I_6$)



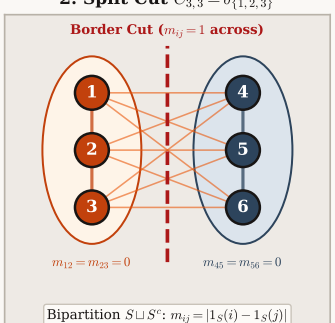
Every pair $i \neq j$ has edge $m_{ij} = 1$ ($\phi = t$)
$\langle D_N c, c \rangle = -\|c\|_2^2 < 0$ on $1^\perp$ (CNSD)

Visualizing Cut Metrics ($C_{k,N-k}$), Tree Metrics, and ℓ^1 Geometry

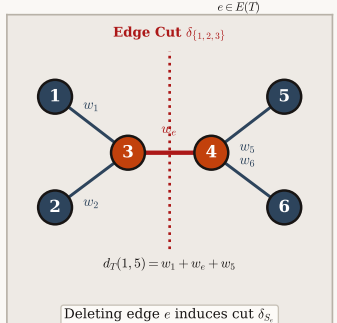
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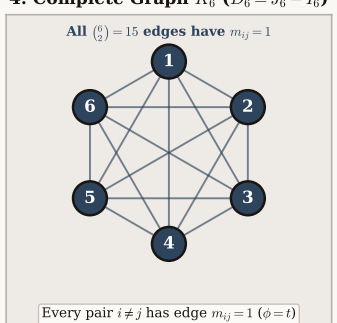
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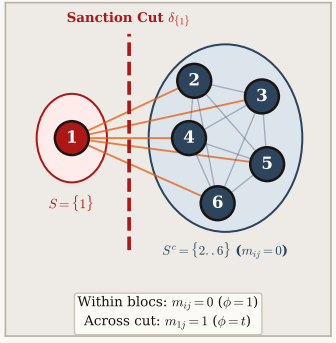


How Cut Pseudo-Metrics and Tree Metrics Operate on a Graph

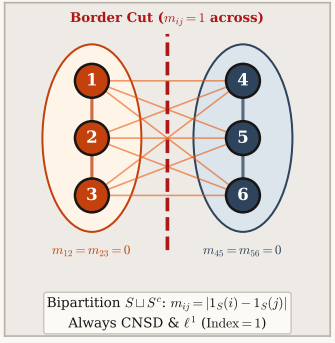
- **Cut Metrics** $C_{k,N-k} = \delta_S$**:** Partition $\{1, \ldots, N\} = S \sqcup S^c$ across a tariff or sanction wall. Nodes inside the *same* bloc trade frictionlessly ($m_{ij} = 0 \iff \phi_{ij} = 1$), while crossing the wall costs $m_{ij} = 1$ ($\phi_{ij} = t$).

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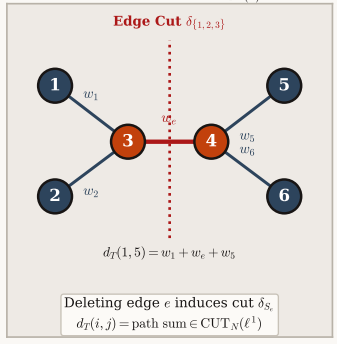
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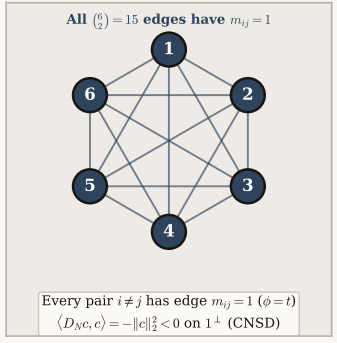
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- **Weighted Trees** ($d_T \in \mathrm{CUT}_N$) **& Universal Barrier** ($D_N = d_{K_N}$)**:** Deleting any edge e of a tree T splits V into two sides $S_e \sqcup S_e^c$, so tree shortest-path distance $d_T = \sum_{e \in E(T)} w_e \delta_{S_e}$ is a positive sum of cuts (ℓ^1-embeddable and unconditionally MT-stable). Likewise, the complete graph K_N ($m_{ij} = 1$ for $i \neq j$) satisfies $\langle D_N c, c \rangle = -\|c\|_2^2 < 0$ on $\mathbf{1}^\perp$.

Does Mossay & Tabuchi's 3-country uniqueness theorem extend to $N = 4$ countries? **Yes!**

Lemma (Every 4-Point Metric Space Lies in the Cut Cone: $\mathrm{MET}_4 = \mathrm{CUT}_4$)

Let $M = (m_{ij})_{i,j=1}^{4} \in \mathrm{MET}_4$ be any pseudo-metric on 4 points. Then M is a non-negative linear combination of the 7 cut metrics on $\{1,2,3,4\}$ (4 singleton cuts $\delta_{\{i\}}$ of type $C_{1,3}$ and 3 pair cuts $\delta_{\{1,2\}}, \delta_{\{1,3\}}, \delta_{\{1,4\}}$ of type $C_{2,2}$):

$$M = \sum_{i=1}^{4} \alpha_i \, \delta_{\{i\}} + \beta_{12} \, \delta_{\{1,2\}} + \beta_{13} \, \delta_{\{1,3\}} + \beta_{14} \, \delta_{\{1,4\}}, \qquad \alpha_i, \beta_{1j} \geq 0.$$

Proof sketch: Relabel $\{1,2,3,4\}$ so that $m_{12} + m_{34} \leq m_{13} + m_{24} \leq m_{14} + m_{23}$. Setting

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Theorem (Unconditional Equilibrium Uniqueness for $N = 4$ Countries)

Since every 4-point metric space is a positive sum of cut metrics ($\mathrm{MET}_4 = \mathrm{CUT}_4 \subset \mathrm{CNSD}_4$), **every 4×4 Freeness-of-Trade matrix Φ is positive semi-definite and Mossay–Tabuchi stable**. Consequently, **every 4-country Mossay–Tabuchi trade model admits a unique equilibrium** for all $L_i > 0$ and all $\sigma > 1$.

The General Metric Embedding Problem

Given a metric space (X, d_X) and a target host space (Y, d_Y) (such as ℓ^2, ℓ^1, or ℓ^∞), **when does (X, d_X) embed isometrically (or with small bi-Lipschitz distortion $D \geq 1$) into (Y, d_Y)?** While every N-point metric space embeds isometrically into ℓ_N^∞ via the Fréchet map $x \mapsto d_X(x, \cdot)$, isometric embeddability into ℓ^2 (Menger–Schoenberg: $d_X^2 \in \mathrm{CNSD}_N$) or ℓ^1 ($d_X \in \mathrm{CUT}_N$) imposes rigid convex-cone constraints inside the metric cone

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Extreme Rays of MET_N, Edit Distance, and Bi-Lipschitz Distortion ($N \geq 5$)

- **Non-Cut Extreme Rays ($N \geq 5$):** A metric $M \neq 0$ spans an **extreme ray** of MET_N if $M = M_1 + M_2 \implies M_1, M_2 \in \mathbb{R}_{\geq 0} M$. For $N \geq 5$, $\mathrm{CUT}_N \subsetneq \mathrm{CNSD}_N \subsetneq \mathrm{MET}_N$: non-CNSD extreme rays emerge ($K_{2,3}$ at $N = 5$; 265 non-cut rays at $N = 6$).

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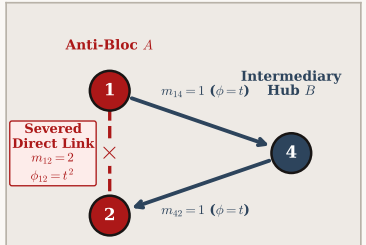
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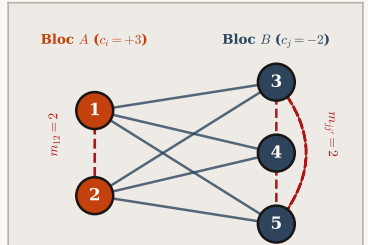
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- **Edit Metric:** The Levenshtein edit distance on strings fails isometric ℓ^1-embeddability (requiring non-constant distortion; Krauthgamer–Rabani 2009) and fails CNSD.

Visualizing Shortest-Path Graph Metrics: How G Induces M_G

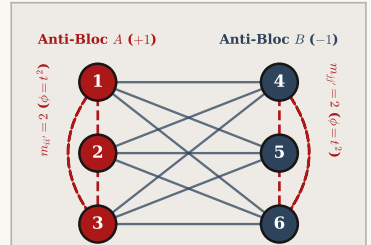
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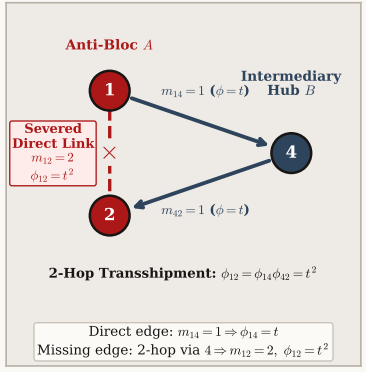


3. Sanctions-Busting Network $K_{3,3}$ ($N = 6$)

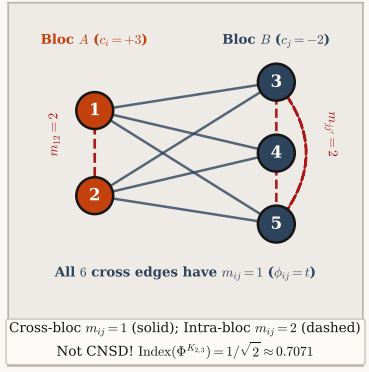


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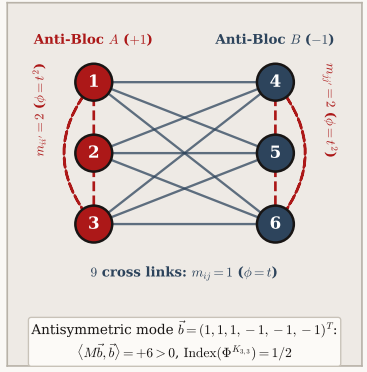
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3. Sanctions-Busting Network $K_{3,3}$ ($N = 6$)

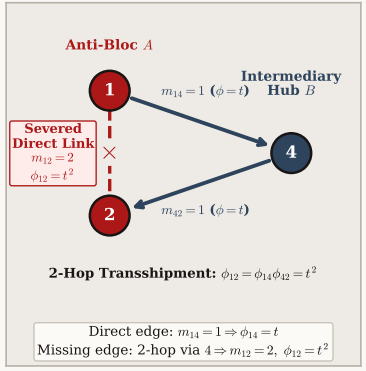


From an Unweighted Trade Network $G = (V, E)$ to the Metric M_G and FoT Matrix Φ_t^G

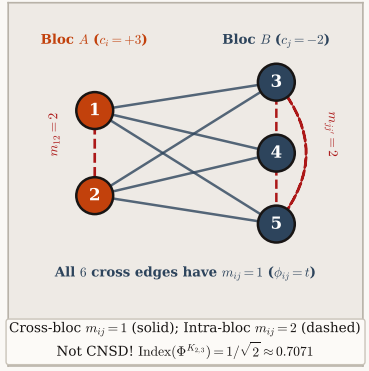
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Visualizing Shortest-Path Graph Metrics: How G Induces M_G

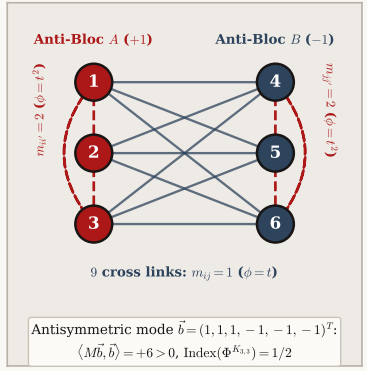
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- **2-Hop Transshipment in Complete Bipartite Graphs $K_{n,m}$:** In $K_{2,3}$ and $K_{3,3}$, states within the same anti-bloc (A or B) have **no direct trade links** ($(i,i') \notin E$) and must route 2 hops through an intermediary k in the opposite bloc ($i \to k \to i'$), yielding $m_{ii'} = 2$ ($\phi_{ii'} = t^2$) and violating CNSD on zero-sum bloc-antisymmetric vectors!

Counterexample to Uniqueness: The $K_{3,3}$ Sanctions-Busting Network

Partition $N = 6$ countries into two "anti-blocs" $A = \{1, 2, 3\}$ (e.g., sanctioning and sanctioned states whose direct trade is severed) and $B = \{4, 5, 6\}$ (isolated sanctions-busting intermediaries).

Example (Shortest-Path Metric on $K_{3,3}$ and Its Freeness-of-Trade Matrix $\Phi_t^{K_{3,3}}$)

Across blocs ($A \times B$), direct trade has distance $m_{ij} = 1$ ($\phi_{ij} = t$). Within A or within B, trade must route through the opposite bloc with distance $m_{ij} = 2$ ($\phi_{ij} = t^2$):

$$M_{K_{3,3}} = \begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 \\ 2 & 0 & 2 & 1 & 1 & 1 \\ 2 & 2 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & 1 & 2 & 2 & 0 \end{pmatrix}, \qquad \Phi_t^{K_{3,3}} = \begin{pmatrix} 1 & t^2 & t^2 & t & t & t \\ t^2 & 1 & t^2 & t & t & t \\ t^2 & t^2 & 1 & t & t & t \\ t & t & t & 1 & t^2 & t^2 \\ t & t & t & t^2 & 1 & t^2 \\ t & t & t & t^2 & t^2 & 1 \end{pmatrix}.$$

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Failure of CNSD and Positive Definiteness for $t \in (\frac{1}{2}, 1)$

Evaluate on the all-ones vector $\vec{a} = (1, 1, 1, 1, 1, 1)^T$ and the zero-sum antisymmetric vector $\vec{b} = (1, 1, 1, -1, -1, -1)^T \in \mathbf{1}^\perp$:

$$\langle M_{K_{3,3}} \vec{b}, \vec{b} \rangle = 6(4 - 3) = +6 > 0 \ \text{(not CNSD!)}, \qquad \Phi_t^{K_{3,3}} \vec{a} = \underbrace{(1 + 2t)(1 + t)}_{\lambda_a(t) > 0} \vec{a}, \qquad \Phi_t^{K_{3,3}} \vec{b} = \underbrace{(1 - 2t)(1 - t)}_{\lambda_b(t)} \vec{b}.$$

For every $t \in (\frac{1}{2}, 1)$, the eigenvalue $\lambda_b(t) = (1 - 2t)(1 - t) < 0$ is **strictly negative**!

Theorem 4.3 (Sufficient Spectral Condition for Multiple Equilibria)

Suppose ΦL has $\vec{a} = (1, \ldots, 1)^T$ and $\vec{b} \in \{-1, 1\}^N$ as eigenvectors with eigenvalues $\lambda_a > 0$ and $\lambda_b < 0$. Whenever

$$\varepsilon \frac{\lambda_b}{\lambda_a} < -1,$$

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Proof & Exact Reduction to Three Equilibria for $K_{3,3}$ ($L_i = 1$)

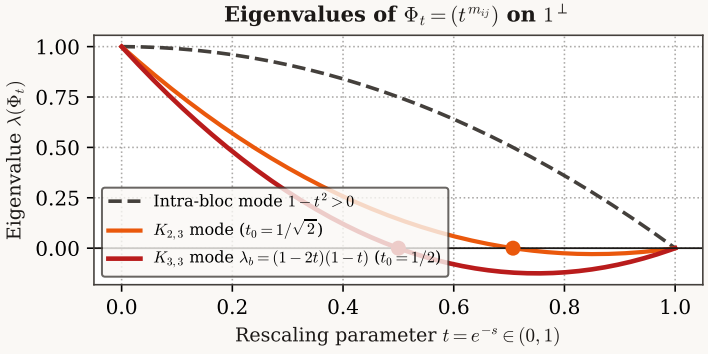
Seek equilibria on the 2-dimensional invariant subspace $v = x\vec{a} + y\vec{b}$ $(x > |y|)$. Since $v^{-\varepsilon} = \frac{(x+y)^{-\varepsilon} + (x-y)^{-\varepsilon}}{2}\vec{a} + \frac{(x+y)^{-\varepsilon} - (x-y)^{-\varepsilon}}{2}\vec{b}$, setting $z := y/x \in (-1, 1)$ and $\tau := \lambda_b/\lambda_a < 0$ reduces $F(v) = 0$ to the single scalar equation

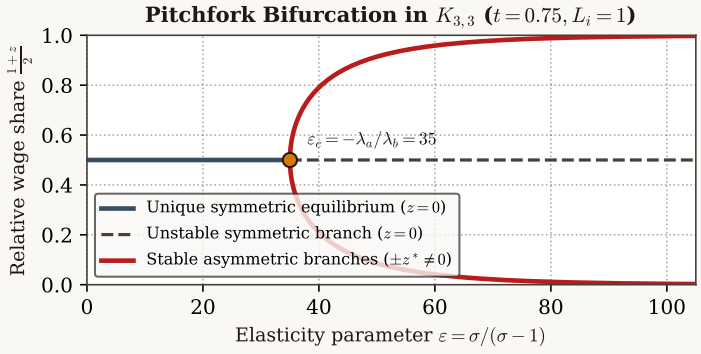
$$z = -\tau \frac{(1-z)^{\varepsilon} - (1+z)^{\varepsilon}}{(1-z)^{\varepsilon} + (1+z)^{\varepsilon}} =: g_{\varepsilon}(z), \qquad \text{where } \tau = \frac{(1-2t)(1-t)}{(1+2t)(1+t)}.$$

Observe that $g_{\varepsilon}(0) = 0$ (the symmetric equilibrium), $\lim_{z \to 1^-} g_{\varepsilon}(z) = -\tau \in (0, 1)$, and $g'_{\varepsilon}(0) = -\varepsilon\tau = -\varepsilon \frac{\lambda_b}{\lambda_a}$. Whenever $-\varepsilon\tau > 1$, i.e.,

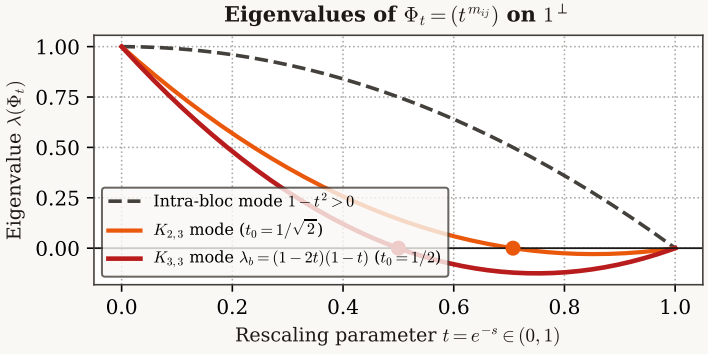
$$\varepsilon > \varepsilon_c(t) := \frac{(1+2t)(1+t)}{(2t-1)(1-t)} \qquad \left(\frac{1}{2} < t < 1\right),$$

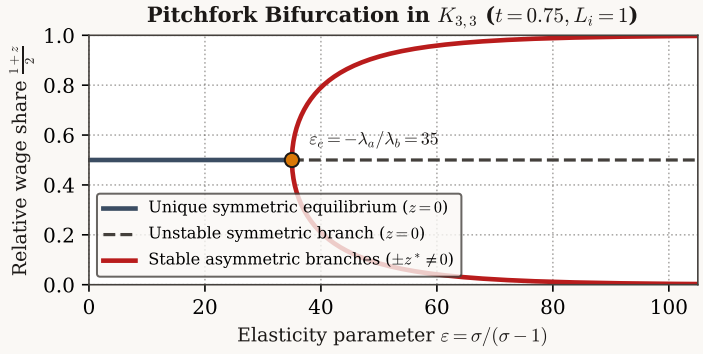
$g_{\varepsilon}(z) = z$ has **three distinct roots**: $z = 0$ (unstable symmetric equilibrium) and $\pm z^* \neq 0$ (two stable **perverse equilibria** where $w \to (1, 1, 1, 0, 0, 0)$ or $(0, 0, 0, 1, 1, 1)$ as $\varepsilon \to \infty$, concentrating all rents in one anti-bloc). ■


Eigenvalues of $\Phi_t = (t^{m_{ij}})$ on $1^\perp$
Eigenvalue $\lambda(\Phi_t)$
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
Intra-bloc mode $1-t^2>0$
$K_{2,3}$ mode ($t_0 = 1/\sqrt{2}$)
$K_{3,3}$ mode $\lambda_b = (1-2t)(1-t)$ ($t_0 = 1/2$)
Rescaling parameter $t = e^{-s} \in (0,1)$


Pitchfork Bifurcation in $K_{3,3}$ ($t = 0.75, L_i = 1$)
Relative wage share $\frac{1+z}{2}$
1.0
0.8
0.6
0.4
0.2
0.0
0
20
40
60
80
100
$\varepsilon_c = -\lambda_a/\lambda_b = 35$
Unique symmetric equilibrium ($z = 0$)
Unstable symmetric branch ($z = 0$)
Stable asymmetric branches ($\pm z^* \neq 0$)
Elasticity parameter $\varepsilon = \sigma/(\sigma-1)$

Spectral Trajectories & Supercritical Pitchfork Bifurcation

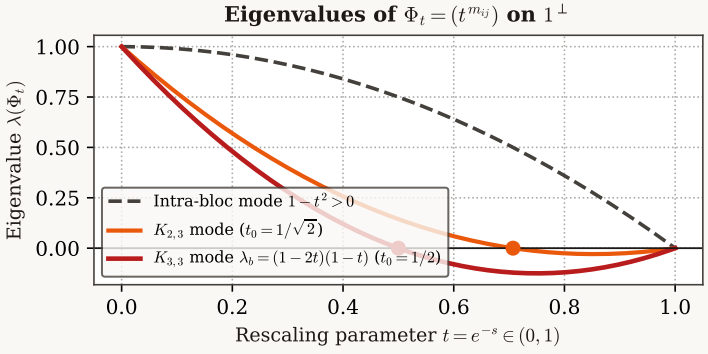


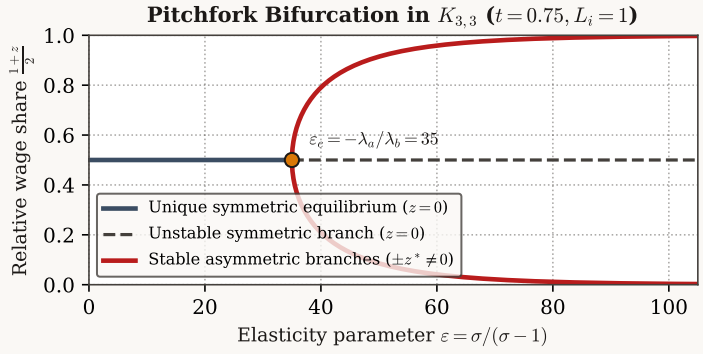


Additional Examples of $K_{n,m}$ "Geometries of Inconvenience"

- **Colonial Mercantilist Hub-and-Spoke Empires:** Colonies barred by Navigation Acts from trading directly with neighboring colonies, forced to route goods through imperial metropole hubs.

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Additional Examples of $K_{n,m}$ "Geometries of Inconvenience"

- **Colonial Mercantilist Hub-and-Spoke Empires:** Colonies barred by Navigation Acts from trading directly with neighboring colonies, forced to route goods through imperial metropole hubs.
- **Landlocked Developing States & Maritime Chokepoints:** Landlocked countries dependent on isolated coastal transit neighbors that do not trade with one another.

The Mossay–Tabuchi Stability Index (Theorem 4.2)

As $\varepsilon \to 0^+$ (high substitutability $\sigma \to \infty$) or $t \to 0^+$ (high friction $\Phi_t \to I_N \succ 0$), the Hessian $\nabla^2 V(u) = \Phi_t + \frac{1}{\varepsilon} D(u) \succ 0$ is always positive definite, guaranteeing a unique equilibrium. We can therefore measure **how fragile a trade network is** by its critical threshold t_0:

Definition (Mossay–Tabuchi Stability Index)

For an $N \times N$ Freeness-of-Trade matrix Φ with friction metric $M_\Phi = (m_{ij})$, the **Mossay–Tabuchi Stability Index** is

$$\text{Index}(\Phi) := \sup \left\{ t_0 \in (0,1] : \Phi_t = (t^{m_{ij}})_{i,j=1}^N \succeq 0 \text{ for all } t \in (0, t_0] \right\}.$$

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Theorem 4.2 (Exact Stability Index of Complete Bipartite Networks $K_{n,m}$)

For the complete bipartite graph $K_{n,m}$ with $n, m \geq 2$,

$$\text{Index}\left(\Phi^{K_{n,m}}\right) = \frac{1}{\sqrt{(n-1)(m-1)}}.$$

Proof: On intra-bloc zero-sum vectors, $\Phi_t^{K_{n,m}}$ acts as $(1-t^2)I \succ 0$. On the 2-dimensional quotient space of bloc-constant vectors, $\Phi_t^{K_{n,m}}$ acts via $Q_t = \begin{pmatrix} 1+(n-1)t^2 & mt \\ nt & 1+(m-1)t^2 \end{pmatrix}$, whose determinant factors identically as

$$\det(Q_t) = \left(1+(n-1)t^2\right)\left(1+(m-1)t^2\right) - nmt^2 = (1-t^2)\left(1-(n-1)(m-1)t^2\right).$$

For $t \in (0,1)$, $\det(Q_t) \geq 0 \iff 1-(n-1)(m-1)t^2 \geq 0 \iff t \leq [(n-1)(m-1)]^{-1/2}$. ■

Why do classical geographical distance models and regional trade blocs never exhibit multiple equilibria?

Theorem (Fractional Powers & Geographical Metrics; Schoenberg, 1938)

If $M = (m_{ij})$ is CNSD, then its entrywise fractional power $M^{\circ\alpha} := (m_{ij}^{\alpha})$ is also CNSD for every $0 < \alpha \leq 1$.

- **Euclidean & Spherical Geography:** For any points $x_1, \ldots, x_N \in \mathbb{R}^d$, $m_{ij} = \|x_i - x_j\|_2^2$ is CNSD, so Euclidean distance $\|x_i - x_j\|_2^{\alpha}$ $(0 < \alpha \leq 2)$ and geodesic distance on the sphere S^2 are CNSD and **unconditionally MT-stable**.

Special Metric Models: "Tectonic Plates," Universal Barriers & Geography

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1. **Tectonic Plates / Regional Border Cuts:** If countries lie on regional plates $S_1, \ldots, S_k$ where crossing boundary S_r incurs tariff $w_r \geq 0$, then $M = \sum_{r=1}^{k} w_r \delta_{S_r} \in \mathrm{CUT}_N \subset \mathrm{CNSD}_N$ is always CNSD and MT-stable.

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2. **Stabilization by Universal Barriers ($D_N = J_N - I_N$):** The discrete metric D_N ($d_{ij} = 1$ for $i \neq j$) satisfies $\langle D_N c, c\rangle = -\|c\|_2^2 < 0$ on $\mathbf{1}^{\perp}$. Hence for **any** metric M, adding a uniform global tariff or fuel shock $M + \alpha D_N$ restores CNSD for all $\alpha \geq \lambda_{\max}(PMP)$. Conversely, uniform trade liberalization ($\alpha \downarrow 0$) in a partitioned world can trigger perverse equilibria (*globalization backlash*).

In quantitative tariff analysis (e.g., evaluating the U.S.–China tariff war), **Allen, Arkolakis (Yale) & Takahashi (2020, *JPE*)** proposed the **Universal Gravity** framework:

The Two-Parameter "Universal Gravity" Tariff Model (Allen, Arkolakis & Takahashi, 2020)

Assuming CES aggregate demand elasticity φ and aggregate supply elasticity ψ (the two scalar *"gravity constants"*), equilibrium output prices p_i and price indices P_i satisfy

$$p_i^{1+\varphi} C_i \left(\frac{p_i}{P_i}\right)^{\psi} = \sum_{j=1}^{N} \tau_{ij}^{-\varphi} P_j^{\varphi} p_j C_j \left(\frac{p_j}{P_j}\right)^{\psi}, \qquad P_i^{-\varphi} = \sum_{j=1}^{N} \tau_{ji}^{-\varphi} p_j^{-\varphi}.$$

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Why Two Scalar Parameters Cannot Replace the Spectral Geometry of Φ

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"Universal Gravity" (Allen, Arkolakis & Takahashi, 2020) vs. Geometry

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2. **Network Geometry Governs Tariff Counterfactuals:** When $\varepsilon > 1$, **two scalar parameters** (φ, ψ) **do not determine equilibrium uniqueness or tariff responses**: if $M_\Phi \in \mathrm{CNSD}_N$, $\Phi_t \succeq 0$ guarantees uniqueness for *all* $\varepsilon > 0$ (Theorem 4.1), whereas if $M_\Phi \notin \mathrm{CNSD}_N$ (e.g., bipartite tariff/sanctions networks $K_{n,m}$), tariff shocks trigger **pitchfork bifurcations into multiple perverse equilibria** (Theorem 4.3)!

Spectral Theory of Metrics (cf. Steinerberger, 2022)

Rather than studying the adjacency matrix of a graph, we study the spectrum of the distance matrix M_Φ on $\mathbf{1}^\perp$ and the kernel family $\Phi_t = (t^{m_{ij}})_{t\in(0,1)}$. Near $t = 1^-$ ($s = -\log t \to 0^+$), $\Phi_t = J_N - sM_\Phi + O(s^2)$, so negative eigenvalues of Φ_t directly correspond to positive eigenvalues of $PM_\Phi P$ (non-CNSD directions).

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Open Problem (The General Bifurcation Problem: Do We Always Get This Bifurcation?)

For a general non-CNSD metric M_Φ (where $\lambda_{\min}(\Phi_t) < 0$ for $t \in (\mathrm{Index}(\Phi), 1)$), when does the equilibrium system $F(v) = 0$ actually bifurcate into multiple equilibria as ε increases? In general, this remains an ***open problem****.*

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What We Solved: The Complete Bipartite Family $K_{n,m}$ (Theorems 4.2 & 4.3)

1. **Theorem 4.3 (± 1 Eigenvector Criterion):** We proved non-uniqueness whenever ΦL admits $\vec{a} = \mathbf{1}$ and a sign vector $\vec{b} \in \{-1, 1\}^N$ as eigenvectors with $\varepsilon\lambda_b/\lambda_a < -1$. For $K_{n,m}$ $(n, m \geq 2)$, this holds beyond $t_0 = [(n-1)(m-1)]^{-1/2}$ (Theorem 4.2) when L balances the two blocs (e.g., $L_i = 1$ for $K_{m,m}$).

Spectral Theory of Metrics (cf. Steinerberger, 2022)

Rather than studying the adjacency matrix of a graph, we study the spectrum of the distance matrix M_Φ on $\mathbf{1}^\perp$ and the kernel family $\Phi_t = (t^{m_{ij}})_{t\in(0,1)}$. Near $t = 1^-$ ($s = -\log t \to 0^+$), $\Phi_t = J_N - sM_\Phi + O(s^2)$, so negative eigenvalues of Φ_t directly correspond to positive eigenvalues of $PM_\Phi P$ (non-CNSD directions).

Open Problem (The General Bifurcation Problem: Do We Always Get This Bifurcation?)

For a general non-CNSD metric M_Φ (where $\lambda_{\min}(\Phi_t) < 0$ for $t \in (\mathrm{Index}(\Phi), 1)$), when does the equilibrium system $F(v) = 0$ actually bifurcate into multiple equilibria as ε increases? In general, this remains an ***open problem****.*

What We Solved: The Complete Bipartite Family $K_{n,m}$ (Theorems 4.2 & 4.3)

1. **Theorem 4.3 (± 1 Eigenvector Criterion):** We proved non-uniqueness whenever ΦL admits $\vec{a} = \mathbf{1}$ and a sign vector $\vec{b} \in \{-1, 1\}^N$ as eigenvectors with $\varepsilon\lambda_b/\lambda_a < -1$. For $K_{n,m}$ $(n, m \geq 2)$, this holds beyond $t_0 = [(n-1)(m-1)]^{-1/2}$ (Theorem 4.2) when L balances the two blocs (e.g., $L_i = 1$ for $K_{m,m}$).
2. **Subtlety Even Within $K_{n,m}$ (Equal $L_i = 1/N$):** When $L_i = 1/N$ on asymmetric $K_{n,m}$, ΦL does not have $\mathbf{1}$ as an eigenvector; the quotient polynomial $P(s) = (1+(m-1)t^2)s^{\varepsilon+1} - mts^\varepsilon + nts - (1+(n-1)t^2) = 0$ bifurcates for large ε **iff** $n, m \geq 3$, whereas $K_{2,3}$ and $K_{2,4}$ retain a **unique equilibrium** for equal L_i!

We visualize policy interventions on $N = 6$ countries as barycentric slices $\mathrm{Conv}(M_1, M_2, M_3) \subset \mathrm{MET}_6$, coloring **MT-stable (CNSD) metrics in slate-smoke** and **non-stable metrics in blaze-red**:

Fig. 3: Cut from Small Bloc

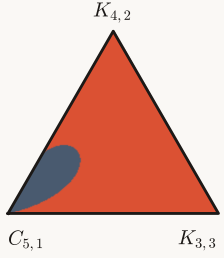


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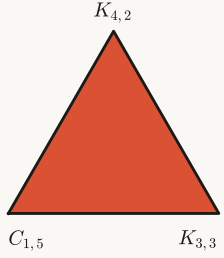


Fig. 5: Universal Barrier D_6

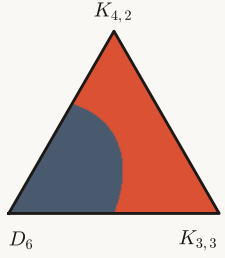
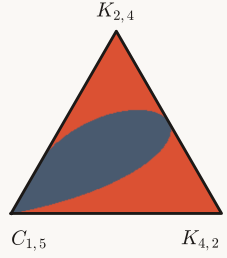


Fig. 6: Cut Placement



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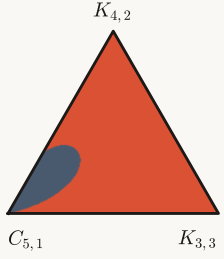


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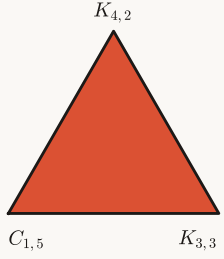


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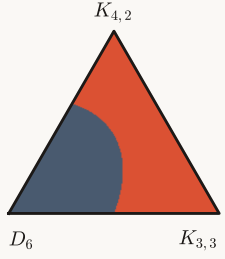
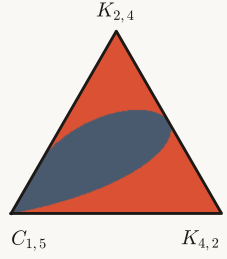


Fig. 6: Cut Placement



Four Policy Lessons from the Metric Cone

- **Fig. 3** $\mathrm{Conv}(C_{5,1}, K_{4,2}, K_{3,3})$**:** Isolating 1 state from the *small* 2-bloc via cut $C_{5,1}$ reduces $K_{4,2} \to K_{4,1}$ (a tree metric in ℓ^1, stabilizing the left edge), whereas $K_{3,3} \to K_{3,2}$ remains unstable.

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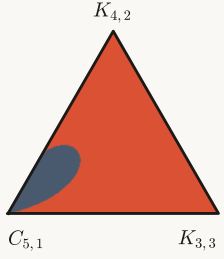


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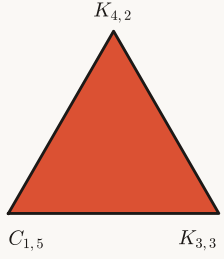


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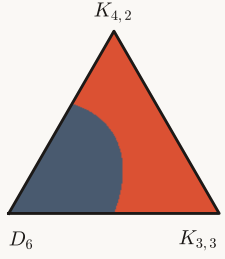
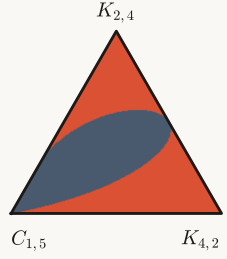


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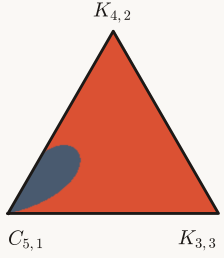


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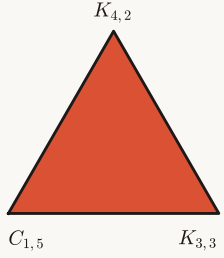


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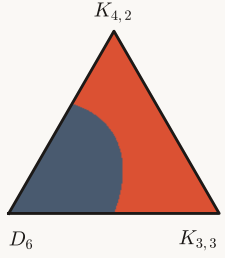
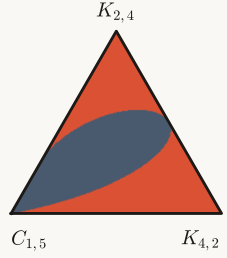


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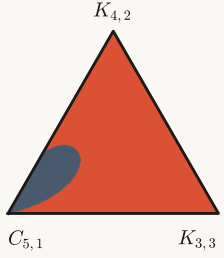


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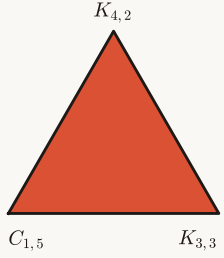


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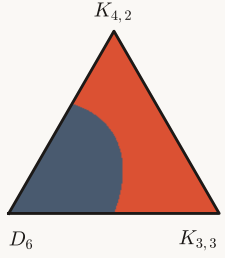
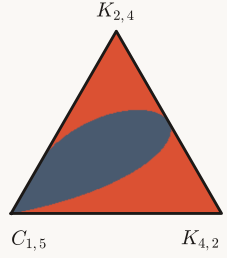


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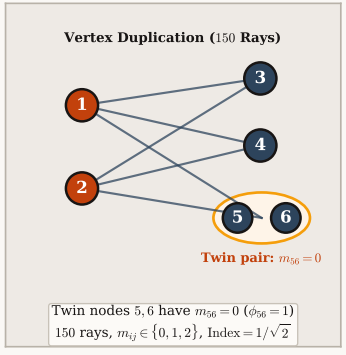


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- **Fig. 6** $\mathrm{Conv}(C_{1,5}, K_{2,4}, K_{4,2})$**:** Targeting the small bottleneck bloc ($K_{2,4} + C_{1,5}$) is pivotal; targeting the large bloc ($K_{4,2} + C_{1,5}$) fails.

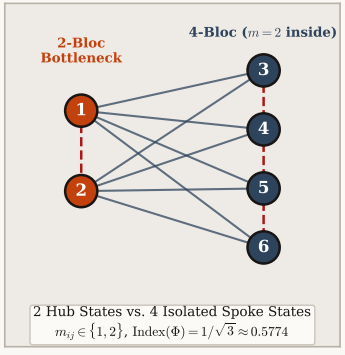
Visual Gallery of the Non-Cut Extreme Ray Graphs of MET_6

Orbits 4–5: Duplicated $K_{2,3}$



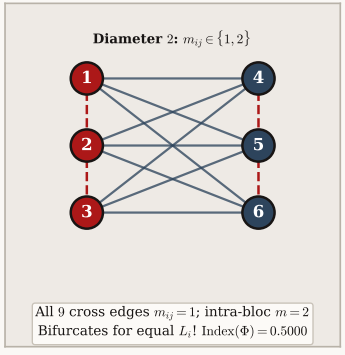
Twin nodes $5,6$ have $m_{56}=0$ ($\phi_{56}=1$)
150 rays, $m_{ij}\in\{0,1,2\}$, Index $=1/\sqrt{2}$

Orbit 6: Bottleneck $K_{2,4}$ (15 Rays)



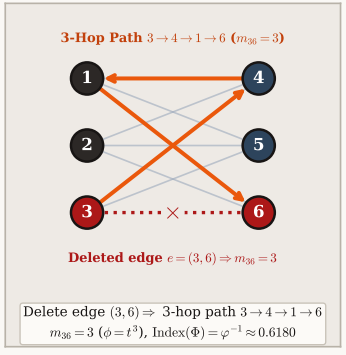
2 Hub States vs. 4 Isolated Spoke States
$m_{ij}\in\{1,2\}$, $\mathrm{Index}(\Phi)=1/\sqrt{3}\approx 0.5774$

Orbit 7: Symmetric $K_{3,3}$ (10 Rays)



All 9 cross edges $m_{ij}=1$; intra-bloc $m=2$
Bifurcates for equal L_i! $\mathrm{Index}(\Phi)=0.5000$

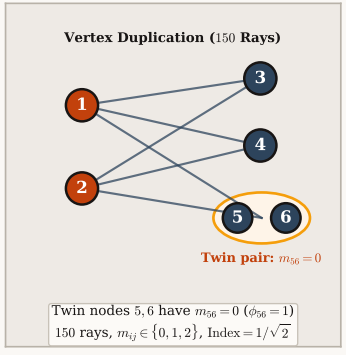
Orbit 8: Twisted $K_{3,3}\setminus e$ (90 Rays)



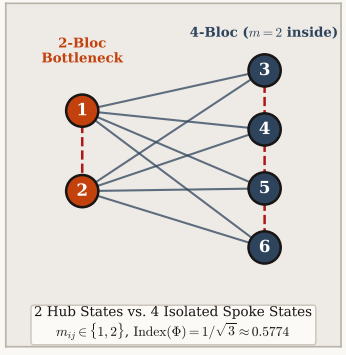
Delete edge $(3,6)\Rightarrow$ 3-hop path $3\to4\to1\to6$
$m_{36}=3$ ($\phi=t^3$), $\mathrm{Index}(\Phi)=\varphi^{-1}\approx 0.6180$

Visual Gallery of the Non-Cut Extreme Ray Graphs of MET_6

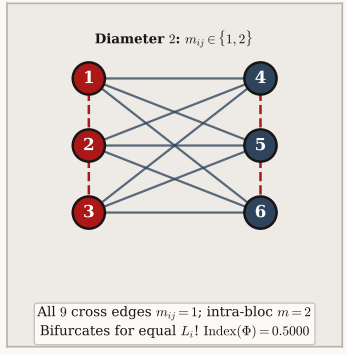
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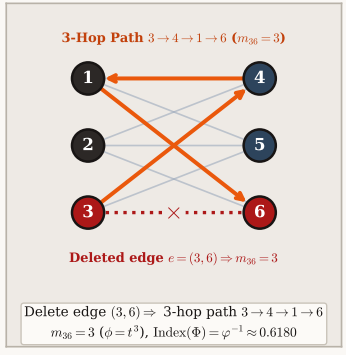
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Orbit 7: Symmetric $K_{3,3}$ (10 Rays)



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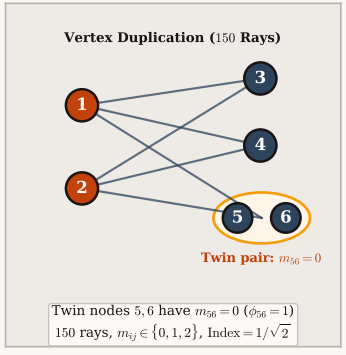


How Graph Topology Generates All 265 Non-Cut Extreme Rays of MET_6 (Orbits 4–8)

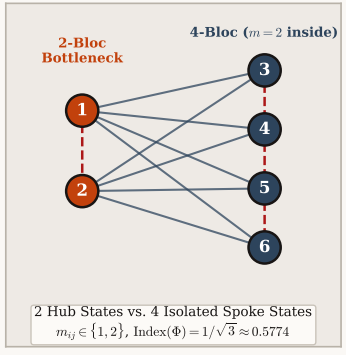
- **Vertex Duplication (Orbits 4–5, 150 Rays) & Bipartite $K_{n,m}$ (Orbits 6–7, 25 Rays):** Duplicating a node in $K_{2,3}$ creates a frictionless twin pair ($m_{56}=0, \phi_{56}=1$) while preserving Index$(\Phi)=1/\sqrt{2}$; $K_{2,4}$ and $K_{3,3}$ have distances $\{1,2\}$ with Index$(\Phi)=[(n-1)(m-1)]^{-1/2}$.

Visual Gallery of the Non-Cut Extreme Ray Graphs of MET_6

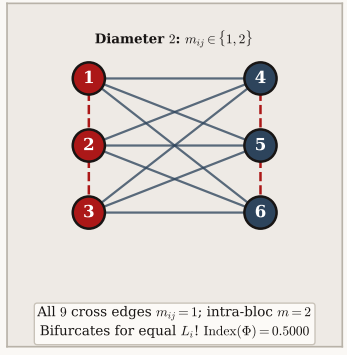
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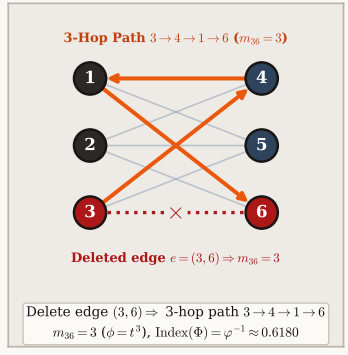
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- **Edge Deletion in $K_{3,3}\setminus e$ (Orbit 8, 90 Rays):** Severing a single cross-bloc link $e=(3,6)$ in $K_{3,3}$ forces trade between nodes $3\in A$ and $6\in B$ along a **3-hop zig-zag path** ($3\to4\to1\to6$), creating distance $m_{36}=3$ ($\phi_{36}=t^3$) and shifting Index(Φ) to the Golden Ratio conjugate $\varphi^{-1}=\frac{\sqrt{5}-1}{2}\approx 0.6180$.

Extreme Rays of MET_5 and MET_6: The Golden Ratio Orbit $K_{3,3} \setminus e$

By exact Double Description enumeration, MET_5 has 25 extreme rays in 3 S_5-orbits ($C_{1,4}, C_{2,3}, K_{2,3}$), and MET_6 has 296 extreme rays in 8 S_6-orbits. Orbit 8 of MET_6 (90 rays) is the shortest-path metric on $K_{3,3} \setminus e$ (deleting edge $(3, 6)$ from $K_{3,3}$, so $m_{36} = 3$):

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Proposition (Golden Ratio Conjugate Stability Index of $K_{3,3} \setminus e$)

The distance matrix $M_{K_{3,3}\setminus e}$ and the antisymmetric block B_t of $\Phi_t^{K_{3,3}\setminus e}$ on $\mathrm{span}\{(1,1,0,-1,-1,0)^T, (0,0,1,0,0,-1)^T\}$ are given by

$$M_{K_{3,3}\setminus e} = \begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 \\ 2 & 0 & 2 & 1 & 1 & 1 \\ 2 & 2 & 0 & 1 & 1 & \mathbf{3} \\ 1 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & \mathbf{3} & 2 & 2 & 0 \end{pmatrix}, \qquad B_t = \begin{pmatrix} 1+t^2-2t & t^2-t \\ 2(t^2-t) & 1-t^3 \end{pmatrix} = (1-t)\begin{pmatrix} 1-t & -t \\ -2t & 1+t+t^2 \end{pmatrix}.$$

By exact Double Description enumeration, MET_5 has 25 extreme rays in 3 S_5-orbits ($C_{1,4}, C_{2,3}, K_{2,3}$), and MET_6 has 296 extreme rays in 8 S_6-orbits. Orbit 8 of MET_6 (90 rays) is the shortest-path metric on $K_{3,3} \setminus e$ (deleting edge $(3,6)$ from $K_{3,3}$, so $m_{36} = 3$):

Proposition (Golden Ratio Conjugate Stability Index of $K_{3,3} \setminus e$)

The distance matrix $M_{K_{3,3}\setminus e}$ and the antisymmetric block B_t of $\Phi_t^{K_{3,3}\setminus e}$ on $\mathrm{span}\{(1,1,0,-1,-1,0)^T, (0,0,1,0,0,-1)^T\}$ are given by

$$M_{K_{3,3}\setminus e} = \begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 \\ 2 & 0 & 2 & 1 & 1 & 1 \\ 2 & 2 & 0 & 1 & 1 & \mathbf{3} \\ 1 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & \mathbf{3} & 2 & 2 & 0 \end{pmatrix}, \qquad B_t = \begin{pmatrix} 1+t^2-2t & t^2-t \\ 2(t^2-t) & 1-t^3 \end{pmatrix} = (1-t)\begin{pmatrix} 1-t & -t \\ -2t & 1+t+t^2 \end{pmatrix}.$$

Its determinant is $\det(B_t) = (1-t)^2(1+t)(1-t-t^2)$, which vanishes on $(0,1)$ at the **Golden Ratio conjugate**:

$$\mathrm{Index}\left(\Phi^{K_{3,3}\setminus e}\right) = \frac{\sqrt{5}-1}{2} = \varphi^{-1} \approx 0.618034.$$

Because $K_{3,3} \setminus e$ preserves the bloc-swapping involution $A \leftrightarrow B$, it **bifurcates into 3 equilibria even for equal country sizes** $L_i = 1/6$!

Complete Census of Extreme Points & Uniqueness for $N = 5$ and $N = 6$

N	Extreme Ray Orbit Class	Rays	Values $\{m_{ij}\}$	CNSD / ℓ^1?	Exact Index(Φ)	Gen. $L_i > 0$	Equal $L_i = 1/N$
5	Orbit 1: Cut $C_{1,4}$ ($\delta_{\{1\}}$)	5	$\{0, 1\}$	Yes (ℓ^1)	1	**Unique (1)**	**Unique (1)**
5	Orbit 2: Cut $C_{2,3}$ ($\delta_{\{1,2\}}$)	10	$\{0, 1\}$	Yes (ℓ^1)	1	**Unique (1)**	**Unique (1)**
5	Orbit 3: Complete Bipartite $K_{2,3}$	10	$\{1, 2\}$	No	$1/\sqrt{2} \approx 0.7071$	**3 Equilibria**	**Unique (1)**
6	Orbits 1–3: Cuts $C_{1,5}, C_{2,4}, C_{3,3}$	31	$\{0, 1\}$	Yes (ℓ^1)	1	**Unique (1)**	**Unique (1)**
6	Orbit 4: $K_{2,3}$ (2-Bloc Vertex Dup.)	60	$\{0, 1, 2\}$	No	$1/\sqrt{2} \approx 0.7071$	**3 Equilibria**	**Unique (1)**
6	Orbit 5: $K_{2,3}$ (3-Bloc Vertex Dup.)	90	$\{0, 1, 2\}$	No	$1/\sqrt{2} \approx 0.7071$	**3 Equilibria**	**Unique (1)**
6	Orbit 6: Complete Bipartite $K_{2,4}$	15	$\{1, 2\}$	No	$1/\sqrt{3} \approx 0.5774$	**3 Equilibria**	**Unique (1)**
6	Orbit 7: Complete Bipartite $K_{3,3}$	10	$\{1, 2\}$	No	$1/2 = 0.5000$	**3 Equilibria**	**3 Equilibria!**
6	Orbit 8: Twisted $K_{3,3} \setminus e$	90	$\{1, 2, 3\}$	No	$\varphi^{-1} = \frac{\sqrt{5}-1}{2} \approx 0.6180$	**3 Equilibria**	**3 Equilibria!**

Complete Census of Extreme Points & Uniqueness for $N = 5$ and $N = 6$

N	Extreme Ray Orbit Class	Rays	Values $\{m_{ij}\}$	CNSD / ℓ^1?	Exact Index(Φ)	Gen. $L_i > 0$	Equal $L_i = 1/N$
5	Orbit 1: Cut $C_{1,4}$ ($\delta_{\{1\}}$)	5	$\{0, 1\}$	Yes (ℓ^1)	1	**Unique (1)**	**Unique (1)**
5	Orbit 2: Cut $C_{2,3}$ ($\delta_{\{1,2\}}$)	10	$\{0, 1\}$	Yes (ℓ^1)	1	**Unique (1)**	**Unique (1)**
5	Orbit 3: Complete Bipartite $K_{2,3}$	10	$\{1, 2\}$	No	$1/\sqrt{2} \approx 0.7071$	**3 Equilibria**	**Unique (1)**
6	Orbits 1–3: Cuts $C_{1,5}, C_{2,4}, C_{3,3}$	31	$\{0, 1\}$	Yes (ℓ^1)	1	**Unique (1)**	**Unique (1)**
6	Orbit 4: $K_{2,3}$ (2-Bloc Vertex Dup.)	60	$\{0, 1, 2\}$	No	$1/\sqrt{2} \approx 0.7071$	**3 Equilibria**	**Unique (1)**
6	Orbit 5: $K_{2,3}$ (3-Bloc Vertex Dup.)	90	$\{0, 1, 2\}$	No	$1/\sqrt{2} \approx 0.7071$	**3 Equilibria**	**Unique (1)**
6	Orbit 6: Complete Bipartite $K_{2,4}$	15	$\{1, 2\}$	No	$1/\sqrt{3} \approx 0.5774$	**3 Equilibria**	**Unique (1)**
6	Orbit 7: Complete Bipartite $K_{3,3}$	10	$\{1, 2\}$	No	$1/2 = 0.5000$	**3 Equilibria**	**3 Equilibria!**
6	Orbit 8: Twisted $K_{3,3} \setminus e$	90	$\{1, 2, 3\}$	No	$\varphi^{-1} = \frac{\sqrt{5}-1}{2} \approx 0.6180$	**3 Equilibria**	**3 Equilibria!**

Computational Check of the $N = 5$ (25 Rays) and $N = 6$ (296 Rays) Extreme Points

Testing each orbit of MET_5 and MET_6 (which consist of cuts, $K_{n,m}$, duplicated $K_{2,3}$, and $K_{3,3} \setminus e$): all non-CNSD rays bifurcate for bloc-balanced $L_i > 0$, while for equal $L_i = 1/N$ only the 100 bloc-symmetric rays on $N = 6$ ($K_{3,3}$ and $K_{3,3} \setminus e$) bifurcate. For general N, the bifurcation problem remains open.

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- **Collaborators & Community:** Tirthankar Bhattacharyya, Sourav Pal, Chandan Pradhan, Tapesh Yadav, Nitin Tomar, Sujit S. Damase, V. B. Kiran Kumar, and an inspiring generation of early-career analysts.

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